

UTKARSH DUDEJA

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SUMMARY

Applied Research Engineer with 2.6+ years turning machine learning research into production-grade systems. I work across the full lifecycle—from prototyping and experimentation to optimizing and deploying models at scale—with deep hands-on experience in distributed data processing and GPU-accelerated computing. An IEEE-published researcher with strong mathematical foundations, I'm drawn to problems where rigorous research meets real-world deployment. Currently seeking applied research and AI/ML engineering roles.

SKILLS

ML & Deep Learning | PyTorch (DDP), TensorFlow, Keras, Scikit-learn, XGBoost, LSTM, CNN, Transformers, Anomaly Detection, Time-Series Modeling
Advanced ML | Continual Learning, Causal & Counterfactual Inference, Explainable AI, Model Compression (Quantization, Pruning)
NLP & GenAI | LangChain, LLMs, Hugging Face, RAG, Vector Databases, Sequence Tagging, Text Classification, LoRA, QLoRA, Agent Architectures
MLOps & Infrastructure | Docker, Kubernetes, PySpark, Ollama, Redis, OpenSearch, CI/CD Pipelines, Cassandra, Linux, Github
Data Engineering | PostgreSQL, Pyspark, ETL, Stream & Batch Processing Pipelines, FastAPI, GraphQL, Event-driven Architecture, Distributed Systems
Languages | Python (Advanced), SQL, Rust (Basic), Golang (Basic)
Networking & Protocols | TCP/IP, Modbus, MQTT, OPC-UA, Wireshark, Suricata, Zeek

PROFESSIONAL EXPERIENCE

WhizHack Technologies

Associate Software Development Engineer

Apr 2024 – Present

1. Research & Development

- Researched and developed a novel multi-layer behavioral anomaly detection system that learns normal network behavior across topology, routine, and register-level physical values autonomously — enabling detection of zero-day threats including unauthorized write, payload inflation and silent device substitution under production workloads: Solution endorsed by CTO.
- Designed a causal inference layer that correlates multi-source anomaly signals into root-caused incidents — combining counterfactual search to quantify deviation from normal, temporal-spatial clustering to link multi-stage attacks, and LLM-driven narration into human-readable form.

2. Production System Engineering

- Engineered a fully automated CVE intelligence pipeline processing 26 years of vulnerability data with 100% ingestion reliability, leveraging NLP classification models to extract structured threat insights from unstructured vulnerability descriptions.
- Re-architected the asset discovery system with a modular, event-driven design, cutting execution time from ~3 hours to ~5 minutes (98% faster) and enabling zero-downtime deployments across 10+ latency-sensitive production environments.
- Architected a distributed, high-performance deep packet analysis system featuring adaptive packet capture, producer-consumer architecture with dynamic worker allocation, and parallel protocol parsing across 15+ industrial protocols, delivering ~60% faster processing in production.
- Optimized and maintained a real-time threat intelligence system with automated multi-source ETL extraction (MITRE, AbuseIPDB, TOR, Feodo, ThreatFox), GraphQL API layer, and concurrent bulk lookup — serving enriched threat data via FastAPI with Redis caching.

3. Distributed ML, Infrastructure & Big Data Processing

- Built a GPU-accelerated distributed anomaly detection system (PySpark + cuML/cuDF) processing network traffic at scale with incremental learning, adaptive clustering, and real-time inference.
- Built the OT lab end-to-end — subnet design, switch configuration, DHCP conflict resolution, and OpenWRT/kernel-level routing — and engineered MITRE ATT&CK for ICS attack scenarios for detection validation.

Software Development Intern

Jan 2024 – Mar 2024

- Contributed to the core R&D team focused on developing security systems; gained hands-on experience with OpenSearch, Fluentd, and Kibana for large-scale data analysis and visualization.

PROJECTS

[ATLAS — Autonomous Multi-Source Threat Correlation & Response Agent](#) | Raspberry Pi 5, Ollama, Qwen 3.5, Langchain, CTI

- Built an autonomous SOC triage agent using LangGraph and a locally hosted LLM that correlates Wazuh and Suricata telemetry, enriches indicators with threat intelligence, and drives automated response decisions behind prompt-injection guardrails.

[LLM-Based Question Answering System](#) | LangChain, FAISS, Mistral-7B, MiniLM

- Built an intelligent document QA agent with LangChain and FAISS vector store, implementing semantic chunking and retrieval with MiniLM embeddings and a quantized Mistral-7B LLM for context-aware answers from unstructured text.

PUBLICATIONS

[Decoding Emotions: Emotion Classification from EEG Brain Signals using AI](#) — IEEE Xplore

[Accurate Object Detection for Intelligent Self-Driving Car System for Mountainous Terrains](#) — Forthcoming

EDUCATION & CERTIFICATIONS

B.Tech in CSE (Hons. Specialization in AI & ML) — Amity University, Uttar Pradesh

2020 – 2024

Duolingo English Test - Advanced: CEFR C1 (135/160) [Link](#)